

Santosh Adhikari

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SUMMARY

Computer Vision & ML Engineer with **7+ years** shipping production deep learning systems across neuroscience research and industrial automation. Proven end-to-end: from model design and edge deployment to cross-platform optimization and global laboratory delivery.

TECHNICAL SKILLS

Languages & ML Frameworks: Python, C++, C | PyTorch, TensorFlow, Keras, scikit-learn

NLP: BERT (embeddings for semantic similarity)

Computer Vision & Edge Deployment: OpenCV, DeepLabCut, pose estimation, instance segmentation, multi-object tracking | TensorRT, ONNX, NVIDIA Jetson Nano, CUDA, Arduino Uno, Raspberry Pi

Data, Backend & Tools: NumPy, Pandas, Flask, FastAPI, PostgreSQL, MongoDB, AWS (EC2, S3) | Docker, Kubernetes, Git, CI/CD, Linux

EXPERIENCE

Conduct Science Inc

Sept 2024 – Present

Computer Vision Engineer

Remote, France

- Engineered automated behavioral analysis pipelines using **DeepLabCut** and custom 2D/3D key-point models to quantify rodent and insect behaviors (grooming, stress); applied **SVM, Random Forest, and UMAP** to extract and cluster behavioral patterns from kinematic features.
- Built an end-to-end **auto-annotation and model training pipeline** for instance segmentation and pose estimation, enabling the company to rapidly train and validate custom models for new species and experimental setups without manual re-engineering.
- Deployed real-time inference on **NVIDIA Jetson Nano** and multi-GPU Windows machines; optimized models via **TensorRT and ONNX** for cross-platform Windows deployment, shipping lightweight, lab-ready software to research institutions worldwide.
- Designed multi-species 2D/3D tracking (pose estimation, segmentation, MOT) integrated as live vision triggers within the core **ConductMaze** automated behavioral software framework.
- Developed high-precision kinematic extraction algorithms that eliminated manual annotation bias, contributing directly to the team's **NIH Replication Prize**-winning research infrastructure.

INRIA

Feb 2024 – Sept 2024

Research Intern

Sophia Antipolis, France

- Analyzed extracellular vesicle secretion and transport in zebrafish embryos using statistical trajectory analysis and ML-based population comparison across experimental conditions.

Parlapp

May 2022 – June 2023

Machine Learning Engineer

Remote

- Designed and deployed scalable ML APIs and async background workers powering a social knowledge-marketplace platform connecting users with shared interests.
- Built a query-to-tag retrieval system using **BERT** embeddings for semantic similarity, enabling real-time extraction of top-N relevant tags from thousands of data points.

IKebana Solutions LLC

Nov 2018 – Sept 2022

Senior Computer Vision Engineer

Kathmandu, Nepal

- Designed, trained, and deployed deep learning models for diverse industrial applications: satellite image segmentation, face-recognition attendance systems, multi-object tracking & re-identification, background removal, document OCR, and activity classification.

- Built a CCTV-based vehicle detection and tracking system deployed in production for the **SoftBank 5G** network project.
- Led ML development for an IoT-based **drowning prevention system** (PoolKnight) with real-time swimmer detection and tracking.

EDUCATION

Université de Bourgogne	Dijon, France
<i>M2 — Masters in Health AI</i>	<i>2022 – 2024</i>
<i>M1 — Masters in Computer Vision and Robotics (VIBOT)</i>	<i>2022 – 2023</i>
Tribhuvan University	Kathmandu, Nepal
<i>B.Eng. in Electronics and Communication Engineering</i>	<i>2014 – 2018</i>

SELECTED PROJECTS

Tomato Plant Disease Detection [\[Paper\]](#) [\[Demo\]](#) Built and manually labelled a custom dataset to train deep learning models detecting 5 tomato diseases. Deployed on Raspberry Pi with Pi Camera directly in the field for real-time harvest analysis.

Virtual Try-On 2D [\[Demo\]](#) Built a GAN-based image translation model allowing users to virtually try on garments over their 2D body images.

Urine Strip Analyzer [\[Demo\]](#) Offline Android app (Java, OpenCV) that automatically quantifies 11 urine parameters via on-device colorimetric analysis – zero cloud dependency.

Traffic Density Measurement [\[Demo\]](#) Vehicle detection and tracking model deployed on embedded hardware for real-time adaptive traffic signal control.

PUBLICATIONS

- Adhikari S. et al. “Virtual Try-on using Image Translation.” *arXiv*, 2023. [\[Paper\]](#)
- Adhikari S. et al. “Tomato Plant Diseases Detection System Using Image Processing.” [\[Paper\]](#)

AWARDS & ACHIEVEMENTS

- **NIH Data Sharing Index (S-Index) Challenge** — Prize Winner, National Institutes of Health, 2026 (1 of 3 winning teams, from 7 national finalists) — Conduct Science Inc
- **NIH Replication Prize** — Track 2: Replication Exemplars Winners (\$50,000) (Team) — Conduct Science Inc; kinematic extraction infrastructure
- **2nd Place** — ActInSpace Hackathon, Besançon, 2022
- **President** — Robotics Club ROBOKATH, Tribhuvan University, 2017–2018
- **1st Runner-up** — Hex-Software Competition, 2017
- **Selected** — [Startup Incubation](#), Kathmandu Maker Faire, 2016